

Mathematical Foundations of Machine Learning

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1 Mathematical Language and Notation

1.1 Sets

A set is a collection of distinct mathematical objects. If x belongs to a set A , we write $x \in A$. If x does not belong to A , we write $x \notin A$.

Important set operations are

$$\begin{aligned}A \cup B &= \{x : x \in A \text{ or } x \in B\}, \\A \cap B &= \{x : x \in A \text{ and } x \in B\}, \\A \setminus B &= \{x : x \in A, x \notin B\},\end{aligned}$$

and

$$A^c = \{x : x \notin A\}.$$

The Cartesian product of two sets is

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

The power set $\mathcal{P}(A)$ is the set of all subsets of A .

1.2 Relations

A relation specifies how elements of sets are associated. An equivalence relation is reflexive, symmetric, and transitive. Equivalence relations partition a set into equivalence classes.

An ordering relation allows elements to be compared. Partial orders are reflexive, antisymmetric, and transitive.

1.3 Functions

A function

$$f : A \rightarrow B$$

assigns exactly one element $f(x) \in B$ to every $x \in A$.

The domain is A , the codomain is B , and the image is

$$f(A) = \{f(x) : x \in A\}.$$

A function is injective if

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

It is surjective if every $y \in B$ has at least one $x \in A$ such that

$$f(x) = y.$$

A function is bijective if it is both injective and surjective.

1.4 Sequences

A sequence is an ordered collection

$$(x_1, x_2, \dots).$$

A sequence (x_n) converges to x if

$$\lim_{n \rightarrow \infty} x_n = x.$$

For real sequences, this means that for every $\epsilon > 0$, there exists N such that

$$|x_n - x| < \epsilon$$

whenever $n \geq N$.

1.5 Limits and Continuity

A function f is continuous at x_0 if

$$\lim_{x \rightarrow x_0} f(x) = f(x_0).$$

Continuity is important because many mathematical results depend on the preservation of limits under functions.

1.6 Asymptotic Notation

The statement

$$f(x) = O(g(x))$$

means that there exist constants $C > 0$ and x_0 such that

$$|f(x)| \leq C|g(x)|$$

for all $x \geq x_0$.

The notation

$$f(x) = o(g(x))$$

means

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0.$$

These concepts describe rates of growth and approximation error.

2 Linear Algebra

2.1 Vectors

A vector in $\mathbb{R}^n$ is written as

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Vector addition and scalar multiplication are defined componentwise:

$$x + y = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad cx = \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}.$$

2.2 Vector Spaces

A vector space V over a field $\mathbb{F}$ is a set closed under vector addition and scalar multiplication and satisfying the vector-space axioms.

A collection $\{v_1, \dots, v_k\}$ is linearly independent if

$$\sum_{i=1}^k a_i v_i = 0$$

implies

$$a_1 = \dots = a_k = 0.$$

A set spans V if every $v \in V$ can be expressed as

$$v = \sum_{i=1}^k a_i v_i.$$

A basis is a linearly independent spanning set. The number of basis vectors is the dimension $\dim V$.

2.3 Matrices

A matrix $A \in \mathbb{R}^{m \times n}$ is a rectangular array

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}.$$

Matrix multiplication is defined by

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}.$$

The transpose satisfies

$$(A^T)_{ij} = A_{ji}.$$

For compatible matrices,

$$(AB)^T = B^T A^T.$$

2.4 Linear Transformations

A map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear if

$$T(ax + by) = aT(x) + bT(y).$$

Every linear transformation between finite-dimensional coordinate spaces can be represented by a matrix:

$$T(x) = Ax.$$

2.5 Column Space and Null Space

The column space of A is

$$\text{Col}(A) = \{Ax : x \in \mathbb{R}^n\}.$$

The null space is

$$\text{Null}(A) = \{x : Ax = 0\}.$$

The rank is

$$\text{rank}(A) = \dim \text{Col}(A).$$

The nullity is

$$\text{nullity}(A) = \dim \text{Null}(A).$$

The rank-nullity theorem is

$$\text{rank}(A) + \text{nullity}(A) = n.$$

2.6 Determinants

For a square matrix A , the determinant is denoted by

$$\det(A).$$

A matrix is invertible if and only if

$$\det(A) \neq 0.$$

For square matrices,

$$\det(AB) = \det(A) \det(B).$$

The determinant represents signed volume scaling under the associated linear transformation.

2.7 Inverse Matrices

A matrix A has inverse A^{-1} when

$$AA^{-1} = A^{-1}A = I.$$

For an invertible matrix,

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

In numerical computation, explicitly forming an inverse is usually less desirable than solving the corresponding linear system.

2.8 Inner Products

The Euclidean inner product is

$$\langle x, y \rangle = x^T y = \sum_{i=1}^n x_i y_i.$$

The Euclidean norm is

$$\|x\|_2 = \sqrt{x^T x}.$$

The Cauchy–Schwarz inequality states

$$|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2.$$

Two vectors are orthogonal if

$$\langle x, y \rangle = 0.$$

2.9 Vector Norms

The L_1 vector norm is

$$\|x\|_1 = \sum_i |x_i|.$$

The Euclidean norm is

$$\|x\|_2 = \left(\sum_i |x_i|^2 \right)^{1/2}.$$

The maximum norm is

$$\|x\|_\infty = \max_i |x_i|.$$

More generally,

$$\|x\|_p = \left(\sum_i |x_i|^p \right)^{1/p}.$$

2.10 Orthogonal Projection

The projection of x onto a nonzero vector u is

$$\text{proj}_u(x) = \frac{x^T u}{u^T u} u.$$

If the columns of A are linearly independent, the orthogonal projection onto $\text{Col}(A)$ is

$$P = A(A^T A)^{-1} A^T.$$

The residual

$$r = x - Px$$

is orthogonal to $\text{Col}(A)$.

2.11 Least Squares

Given

$$Ax \approx b,$$

the least-squares problem is

$$\min_x \|Ax - b\|_2^2.$$

The normal equations are

$$A^T A x = A^T b.$$

When $A^T A$ is invertible,

$$x = (A^T A)^{-1} A^T b.$$

2.12 Eigenvalues and Eigenvectors

An eigenvector $v \neq 0$ satisfies

$$Av = \lambda v.$$

The corresponding eigenvalue λ satisfies

$$\det(A - \lambda I) = 0.$$

If a matrix has a complete set of linearly independent eigenvectors,

$$A = V\Lambda V^{-1},$$

where Λ is diagonal.

2.13 Symmetric Matrices

A matrix is symmetric if

$$A = A^T.$$

Every real symmetric matrix has real eigenvalues and an orthonormal eigenbasis. Therefore,

$$A = Q\Lambda Q^T$$

for an orthogonal matrix Q .

2.14 Positive Definite Matrices

A symmetric matrix A is positive definite if

$$x^T Ax > 0$$

for every $x \neq 0$.

It is positive semidefinite if

$$x^T Ax \geq 0.$$

Positive definiteness is closely connected to strict convexity and uniqueness of optimization solutions.

2.15 Quadratic Forms

A quadratic form is

$$q(x) = x^T Ax.$$

For symmetric positive definite A ,

$$q(x) > 0$$

for every nonzero x .

2.16 Matrix Decompositions

LU decomposition expresses a matrix as

$$A = LU,$$

where L is lower triangular and U is upper triangular.

QR decomposition is

$$A = QR,$$

where Q has orthonormal columns and R is upper triangular.

The singular value decomposition is

$$A = U\Sigma V^T.$$

The singular values are the diagonal entries of Σ and are nonnegative.

SVD provides a fundamental description of the geometry, rank, conditioning, and low-rank approximation of a matrix.

3 Multivariable Calculus

3.1 Derivative

For a scalar function $f : \mathbb{R} \rightarrow \mathbb{R}$, the derivative is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

It represents the local rate of change of the function.

3.2 Product and Chain Rules

The product rule is

$$(fg)' = f'g + fg'.$$

The chain rule is

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x).$$

The chain rule is fundamental for differentiating compositions of functions.

3.3 Partial Derivatives

For

$$f : \mathbb{R}^n \rightarrow \mathbb{R},$$

the partial derivative with respect to x_i is

$$\frac{\partial f}{\partial x_i}.$$

The gradient is

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}.$$

3.4 Directional Derivative

The directional derivative of f in direction v is

$$D_v f(x) = \nabla f(x)^T v.$$

For $\|v\|_2 = 1$, the gradient direction gives the maximum directional derivative:

$$\max_{\|v\|_2=1} D_v f(x) = \|\nabla f(x)\|_2.$$

3.5 Jacobian

For

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m,$$

the Jacobian matrix is

$$J_f(x) = \left[\frac{\partial f_i}{\partial x_j} \right]_{i,j}.$$

The first-order approximation is

$$f(x + \Delta x) \approx f(x) + J_f(x)\Delta x.$$

3.6 Hessian

For a scalar function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the Hessian is

$$\nabla^2 f(x) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right]_{i,j}.$$

It describes local second-order curvature.

3.7 Taylor Expansion

The multivariable second-order Taylor approximation is

$$f(x+h) \approx f(x) + \nabla f(x)^T h + \frac{1}{2} h^T \nabla^2 f(x) h.$$

Higher-order derivatives provide higher-order corrections.

3.8 Local Minima

If x^* is an interior local minimum of a differentiable function, then

$$\nabla f(x^*) = 0.$$

If additionally

$$\nabla^2 f(x^*) \succ 0,$$

then x^* is a strict local minimum.

3.9 Matrix Calculus

Important identities include

$$\begin{aligned}\nabla_x (a^T x) &= a, \\ \nabla_x (x^T A x) &= (A + A^T)x,\end{aligned}$$

and if A is symmetric,

$$\nabla_x (x^T A x) = 2Ax.$$

For

$$f(x) = \frac{1}{2} x^T A x - b^T x,$$

with symmetric A ,

$$\nabla f(x) = Ax - b.$$

4 Probability Theory

4.1 Probability Space

A probability space consists of

$$(\Omega, \mathcal{F}, \mathbb{P}),$$

where Ω is the sample space, $\mathcal{F}$ is a sigma-algebra of events, and $\mathbb{P}$ is a probability measure.

The axioms include

$$\mathbb{P}(\Omega) = 1,$$

$$\mathbb{P}(A) \geq 0,$$

and for pairwise disjoint events,

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i).$$

4.2 Conditional Probability

For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

The multiplication rule is

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B).$$

4.3 Bayes' Rule

Bayes' theorem is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

For a parameter θ and data x ,

$$p(\theta \mid x) = \frac{p(x \mid \theta)p(\theta)}{p(x)}.$$

4.4 Independence

Events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Random variables X and Y are independent when

$$p(x, y) = p(x)p(y).$$

4.5 Random Variables

A random variable is a measurable function

$$X : \Omega \rightarrow \mathbb{R}.$$

Its cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x).$$

For a discrete random variable,

$$p_X(x) = \mathbb{P}(X = x).$$

For a continuous random variable, a density f_X satisfies

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

4.6 Expectation

For a discrete random variable,

$$\mathbb{E}[X] = \sum_x xp_X(x).$$

For a continuous random variable,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xf_X(x) dx.$$

Expectation is linear:

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

4.7 Moments

The k th raw moment is

$$\mathbb{E}[X^k].$$

The k th central moment is

$$\mathbb{E}[(X - \mathbb{E}[X])^k].$$

The variance is

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Equivalently,

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

4.8 Covariance

The covariance is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Equivalently,

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The covariance matrix of a random vector X is

$$\Sigma = \mathbb{E}[(X - \mu)(X - \mu)^T],$$

where

$$\mu = \mathbb{E}[X].$$

4.9 Correlation

The correlation coefficient is

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

It satisfies

$$-1 \leq \rho_{XY} \leq 1.$$

4.10 Joint and Marginal Distributions

For a joint density,

$$f_{X,Y}(x, y),$$

the marginal density of X is

$$f_X(x) = \int f_{X,Y}(x, y) dy.$$

The conditional density is

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

4.11 Law of Large Numbers

For iid random variables with finite expectation,

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

converges to $\mathbb{E}[X]$ as $n \rightarrow \infty$ under the appropriate law-of-large-numbers conditions.

This provides the mathematical basis for replacing population expectations with empirical averages.

4.12 Central Limit Theorem

If $X_1, \dots, X_n$ are iid with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2,$$

then

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \Rightarrow \mathcal{N}(0, 1).$$

The CLT explains why normalized averages often approach a Gaussian distribution.

5 Important Probability Distributions

5.1 Bernoulli Distribution

For

$$\begin{aligned} X &\sim \text{Bernoulli}(p), \\ \mathbb{P}(X = 1) &= p, \quad \mathbb{P}(X = 0) = 1 - p. \end{aligned}$$

Its mean and variance are

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p).$$

5.2 Binomial Distribution

For

$$X \sim \text{Binomial}(n, p),$$
$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Its expectation and variance are

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1-p).$$

5.3 Categorical Distribution

A categorical random variable takes values in

$$\{1, \dots, K\}$$

with probabilities

$$p_1, \dots, p_K, \quad \sum_{k=1}^K p_k = 1.$$

5.4 Multinomial Distribution

For counts

$$(N_1, \dots, N_K)$$

with total count n ,

$$\mathbb{P}(N_1 = n_1, \dots, N_K = n_K) = \frac{n!}{\prod_k n_k!} \prod_k p_k^{n_k}.$$

5.5 Uniform Distribution

For

$$X \sim \text{Uniform}(a, b),$$

the density is

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

The mean and variance are

$$\mathbb{E}[X] = \frac{a+b}{2},$$
$$\text{Var}(X) = \frac{(b-a)^2}{12}.$$

5.6 Gaussian Distribution

For

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

the density is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

For a multivariate Gaussian,

$$X \sim \mathcal{N}(\mu, \Sigma),$$

the density is

$$f(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right).$$

5.7 Poisson Distribution

For

$$X \sim \text{Poisson}(\lambda),$$
$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

Its mean and variance are

$$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

5.8 Exponential Distribution

For rate $\lambda > 0$,

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

The mean and variance are

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

6 Mathematical Statistics

6.1 Population and Sample

A population is represented by a probability distribution. A sample consists of observations

$$X_1, \dots, X_n.$$

Under iid sampling,

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_\theta.$$

A statistic is any function of the observations:

$$T = T(X_1, \dots, X_n).$$

6.2 Sample Mean

The sample mean is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

If the observations have mean μ ,

$$\mathbb{E}[\bar{X}] = \mu.$$

If they have variance σ^2 and are independent,

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

6.3 Sample Variance

The unbiased sample variance is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

6.4 Likelihood

Given observations $x_1, \dots, x_n$, the likelihood is

$$L(\theta; x_{1:n}) = \prod_{i=1}^n f(x_i | \theta)$$

under iid sampling.

The log-likelihood is

$$\ell(\theta) = \sum_{i=1}^n \log f(x_i | \theta).$$

6.5 Maximum Likelihood Estimation

The maximum likelihood estimator is

$$\hat{\theta}_{\text{MLE}} =_{\theta} L(\theta; x_{1:n}).$$

Because the logarithm is monotonic,

$$\hat{\theta}_{\text{MLE}} =_{\theta} \ell(\theta).$$

At an interior differentiable optimum,

$$\nabla_{\theta} \ell(\hat{\theta}) = 0.$$

6.6 Maximum A Posteriori Estimation

Bayes' rule gives

$$p(\theta | x) \propto p(x | \theta)p(\theta).$$

The MAP estimator is

$$\hat{\theta}_{\text{MAP}} =_{\theta} p(x | \theta)p(\theta).$$

Equivalently,

$$\hat{\theta}_{\text{MAP}} =_{\theta} [\log p(x | \theta) + \log p(\theta)].$$

6.7 Sufficient Statistics

A statistic $T(X)$ is sufficient for θ if, informally, it contains all information in the sample relevant to inference about θ .

The factorization theorem states that a statistic is sufficient when the likelihood can be written as

$$p(x | \theta) = g(T(x), \theta)h(x).$$

6.8 Bias

The bias of an estimator $\hat{\theta}$ is

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta.$$

An estimator is unbiased when

$$\mathbb{E}[\hat{\theta}] = \theta.$$

6.9 Variance and MSE

The mean squared error is

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2].$$

The bias-variance decomposition is

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

6.10 Fisher Information

For scalar θ , Fisher information is

$$I(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log f(X | \theta) \right)^2 \right].$$

Under regularity conditions,

$$I(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(X | \theta) \right].$$

For independent observations,

$$I_n(\theta) = nI(\theta).$$

7 Information Theory

7.1 Entropy

For a discrete random variable,

$$H(X) = - \sum_x p(x) \log p(x).$$

Entropy measures uncertainty in a probability distribution.

7.2 Joint Entropy

For two random variables,

$$H(X, Y) = - \sum_{x,y} p(x, y) \log p(x, y).$$

7.3 Conditional Entropy

Conditional entropy is

$$H(Y | X) = - \sum_{x,y} p(x, y) \log p(y | x).$$

The chain rule is

$$H(X, Y) = H(X) + H(Y | X).$$

7.4 Cross-Entropy

The cross-entropy between distributions p and q is

$$H(p, q) = - \sum_x p(x) \log q(x).$$

It can be decomposed as

$$H(p, q) = H(p) + D_{\text{KL}}(p \| q).$$

7.5 Kullback–Leibler Divergence

The KL divergence is

$$D_{\text{KL}}(p \| q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

It satisfies

$$D_{\text{KL}}(p \| q) \geq 0.$$

Equality holds when $p = q$ almost everywhere.

KL divergence is generally not symmetric:

$$D_{\text{KL}}(p \| q) \neq D_{\text{KL}}(q \| p).$$

7.6 Mutual Information

Mutual information is

$$I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}.$$

Equivalent forms include

$$I(X; Y) = H(X) - H(X | Y),$$

and

$$I(X; Y) = H(Y) - H(Y | X).$$

It is also equal to

$$I(X; Y) = D_{\text{KL}}(p(x, y) \| p(x)p(y)).$$

Therefore,

$$I(X; Y) \geq 0.$$

Moreover,

$$I(X; Y) = 0$$

if and only if X and Y are independent.

8 Optimization

8.1 Optimization Problems

A general optimization problem is

$$\min_{x \in \mathcal{X}} f(x).$$

A point x^* is a global minimizer if

$$f(x^*) \leq f(x)$$

for every feasible x .

A local minimizer satisfies this inequality only in some neighborhood of x^* .

8.2 Stationary Points

A differentiable unconstrained local optimum satisfies

$$\nabla f(x^*) = 0.$$

A point satisfying this equation is called a stationary point.

A stationary point may be a minimum, maximum, or saddle point.

8.3 First-Order Conditions

For differentiable f , a necessary condition for an interior local minimum is

$$\nabla f(x^*) = 0.$$

For a convex differentiable function, every stationary point is a global minimum.

8.4 Second-Order Conditions

If

$$\nabla f(x^*) = 0$$

and

$$\nabla^2 f(x^*) \succ 0,$$

then x^* is a strict local minimum.

If

$$\nabla^2 f(x^*) \prec 0,$$

then x^* is a strict local maximum.

If the Hessian is indefinite, the point is a saddle point.

8.5 Gradient Descent

Gradient descent uses

$$x_{k+1} = x_k - \eta_k \nabla f(x_k),$$

where $\eta_k > 0$ is the step size.

The gradient satisfies

$$D_v f(x) = \nabla f(x)^T v.$$

Therefore, $-\nabla f(x)$ is the direction of steepest local decrease under the Euclidean norm.

8.6 Newton's Method

Newton's method uses second-order information:

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k).$$

It is derived by solving the local quadratic approximation.

8.7 Line Search

A line-search method chooses a direction d_k and searches for a step size α_k :

$$x_{k+1} = x_k + \alpha_k d_k.$$

A descent direction satisfies

$$\nabla f(x_k)^T d_k < 0.$$

8.8 Stochastic Objectives

Many objectives have the form

$$F(\theta) = \mathbb{E}_\xi[\ell(\theta; \xi)].$$

If

$$g(\theta; \xi)$$

satisfies

$$\mathbb{E}[g(\theta; \xi)] = \nabla F(\theta),$$

then it is an unbiased stochastic gradient estimator.

8.9 Convergence

A sequence (x_k) converges to x^* if

$$\lim_{k \rightarrow \infty} \|x_k - x^*\| = 0.$$

Optimization algorithms can also be studied through

$$f(x_k) \rightarrow f^*$$

or

$$\|\nabla f(x_k)\| \rightarrow 0.$$

These are different notions of convergence.

9 Convex Analysis

9.1 Convex Sets

A set C is convex if

$$tx + (1 - t)y \in C$$

for every $x, y \in C$ and $t \in [0, 1]$.

Thus every line segment joining two points in C remains inside C .

9.2 Convex Functions

A function $f : C \rightarrow \mathbb{R}$ is convex if

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$$

for all $x, y \in C$ and $t \in [0, 1]$.

For differentiable f , convexity implies

$$f(y) \geq f(x) + \nabla f(x)^T (y - x).$$

9.3 Second-Order Characterization

For a twice differentiable function on an open convex domain,

$$\nabla^2 f(x) \succeq 0$$

characterizes convexity.

If

$$\nabla^2 f(x) \succ 0$$

throughout the domain, the function is strictly convex under appropriate conditions.

9.4 Strong Convexity

A function is m -strongly convex if

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{m}{2} \|y - x\|_2^2$$

for $m > 0$.

Strong convexity provides curvature and guarantees uniqueness of a global minimizer.

9.5 Jensen's Inequality

For convex f ,

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

whenever the quantities are well-defined.

For concave functions, the inequality is reversed.

9.6 Convex Optimization

A convex optimization problem has the form

$$\min_x f(x)$$

where f is convex and the feasible set is convex.

The central property is that every local minimum is a global minimum.

10 Constrained Optimization

10.1 Equality Constraints

Consider

$$\min_x f(x)$$

subject to

$$h(x) = 0.$$

The Lagrangian is

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T h(x).$$

The first-order conditions are

$$\begin{aligned} \nabla_x \mathcal{L}(x^*, \lambda^*) &= 0, \\ h(x^*) &= 0. \end{aligned}$$

10.2 Inequality Constraints

Consider

$$\min_x f(x)$$

subject to

$$g_i(x) \leq 0.$$

The Lagrangian is

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_j \lambda_j h_j(x) + \sum_i \mu_i g_i(x).$$

The inequality multipliers satisfy

$$\mu_i \geq 0.$$

10.3 KKT Conditions

The Karush–Kuhn–Tucker conditions consist of:

$$\nabla_x \mathcal{L}(x^*, \lambda^*, \mu^*) = 0,$$

$$h_j(x^*) = 0,$$

$$g_i(x^*) \leq 0,$$

$$\mu_i^* \geq 0,$$

and complementary slackness:

$$\mu_i^* g_i(x^*) = 0.$$

Under appropriate constraint qualifications, these conditions characterize constrained optima for convex problems.

10.4 Duality

The Lagrangian dual function is

$$q(\lambda, \mu) = \inf_x \mathcal{L}(x, \lambda, \mu).$$

The dual problem is

$$\max_{\lambda, \mu} q(\lambda, \mu)$$

subject to

$$\mu \geq 0.$$

Weak duality gives

$$q(\lambda, \mu) \leq p^*,$$

where p^* is the primal optimum.

Under suitable conditions, strong duality gives

$$p^* = d^*.$$

11 Norms, Geometry, and Function Spaces

11.1 Norms

A norm $\|\cdot\|$ satisfies

$$\|x\| \geq 0,$$

$$\|x\| = 0 \iff x = 0,$$

$$\|\alpha x\| = |\alpha| \|x\|,$$

and

$$\|x + y\| \leq \|x\| + \|y\|.$$

11.2 Metrics

Every norm induces a metric:

$$d(x, y) = \|x - y\|.$$

This gives a notion of distance and convergence.

11.3 L_p Spaces

For $1 \leq p < \infty$,

$$\|f\|_p = \left(\int |f(x)|^p dx \right)^{1/p}.$$

The space L_p consists of functions whose p th power is integrable, with functions identified when they are equal almost everywhere.

The infinity norm is denoted by

$$\|f\|_\infty.$$

11.4 Inner-Product Spaces

An inner product induces a norm:

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

Orthogonality is defined by

$$\langle x, y \rangle = 0.$$

Projection onto closed subspaces is a fundamental geometric operation.

11.5 Banach Spaces

A Banach space is a complete normed vector space.

Completeness means every Cauchy sequence converges to an element of the space.

11.6 Hilbert Spaces

A Hilbert space is a complete inner-product space.

Hilbert spaces generalize Euclidean geometry to infinite-dimensional settings.

Important examples include

$$\mathbb{R}^n$$

and suitable L_2 function spaces.

12 Approximation Theory

12.1 Function Approximation

Approximation theory studies how a complicated function can be represented or approximated by functions from a simpler class.

Given target f and approximation g , the approximation error may be measured by

$$\|f - g\|.$$

Different norms correspond to different notions of approximation quality.

12.2 Interpolation

Interpolation constructs a function satisfying prescribed values:

$$p(x_i) = f(x_i).$$

Polynomial interpolation in Lagrange form is

$$p(x) = \sum_{i=0}^n f(x_i) L_i(x),$$

where

$$L_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

12.3 Least-Squares Approximation

Suppose

$$f(x) \approx \sum_{j=1}^m c_j \phi_j(x).$$

The coefficients may be chosen by minimizing

$$\int \left(f(x) - \sum_{j=1}^m c_j \phi_j(x) \right)^2 dx.$$

This is a projection problem in a function space.

12.4 Approximation Error

The total approximation error can be described through

$$\|f - f_m\|.$$

Convergence of an approximation sequence means

$$\|f - f_m\| \rightarrow 0$$

as the approximation dimension m increases.

13 Numerical Mathematics

13.1 Floating-Point Arithmetic

Computers represent real numbers using finite precision. Therefore, arithmetic generally introduces rounding errors:

$$\text{fl}(x) = x + \delta.$$

Numerical algorithms must distinguish mathematical error from floating-point error.

13.2 Numerical Differentiation

The forward difference is

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}.$$

The backward difference is

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}.$$

The centered difference is

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}.$$

13.3 Numerical Integration

The trapezoidal rule approximates

$$\int_a^b f(x) dx$$

by

$$\frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right].$$

13.4 Numerical Linear Algebra

Linear systems are written as

$$Ax = b.$$

Direct methods include Gaussian elimination and matrix factorizations.

Iterative methods construct a sequence

$$x_0, x_1, \dots$$

that approaches a solution.

13.5 Conditioning

The condition number of a matrix under a chosen norm is

$$\kappa(A) = \|A\| \|A^{-1}\|.$$

A large condition number indicates sensitivity of the solution to perturbations in the input.

13.6 Numerical Stability

An algorithm is numerically stable when its computed result remains close to the exact result for a nearby problem.

Stability and conditioning are distinct:

- conditioning describes the sensitivity of the mathematical problem;
- stability describes the behavior of the numerical algorithm.

14 Fourier Analysis

14.1 Fourier Series

A periodic function can be represented, under appropriate conditions, as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)].$$

The coefficients are

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx,$$

and

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx.$$

14.2 Fourier Transform

The continuous Fourier transform is

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx.$$

The inverse transform is

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} d\omega.$$

Fourier analysis converts spatial or temporal information into frequency information.

14.3 Discrete Fourier Transform

For a discrete sequence $x_0, \dots, x_{N-1}$,

$$X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n / N}.$$

The inverse transform is

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{2\pi i k n / N}.$$

14.4 Convolution

The continuous convolution of f and g is

$$(f * g)(x) = \int f(\tau)g(x - \tau) d\tau.$$

The convolution theorem states

$$\widehat{f * g} = \hat{f} \hat{g}.$$

Thus convolution in physical space corresponds to multiplication in frequency space.

14.5 Spectral Representation

Many mathematical operators can be understood through their action on frequency components. For differential operators, Fourier transformation often converts differentiation into multiplication:

$$\widehat{\frac{df}{dx}} = i\omega \hat{f}(\omega).$$

This provides a powerful method for analyzing differential equations and linear systems.

15 Differential Equations

15.1 Ordinary Differential Equations

An ordinary differential equation relates an unknown function to its derivatives:

$$\frac{dy}{dt} = f(t, y).$$

A solution is a function satisfying the equation on an appropriate interval.

15.2 Initial Value Problems

An initial value problem specifies

$$\frac{dy}{dt} = f(t, y),$$

together with

$$y(t_0) = y_0.$$

Under appropriate regularity conditions, existence and uniqueness can be established.

15.3 Partial Differential Equations

A partial differential equation contains partial derivatives of a function of several variables.

A general PDE may be written as

$$F(x, u, \nabla u, \nabla^2 u, \dots) = 0.$$

15.4 Diffusion Equation

The diffusion equation is

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u.$$

It describes diffusion and smoothing processes.

15.5 Wave Equation

The wave equation is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u.$$

It describes propagation of disturbances.

15.6 Boundary Conditions

A PDE problem may require conditions on the boundary of a domain.

Dirichlet conditions specify the function:

$$u = g.$$

Neumann conditions specify the normal derivative:

$$\frac{\partial u}{\partial n} = g.$$

Boundary conditions are essential for well-posedness.

15.7 Variational Formulations

Many differential equations can be represented through minimization of an energy functional:

$$J[u] = \int_{\Omega} F(x, u, \nabla u) dx.$$

The corresponding Euler–Lagrange equation is

$$\frac{\partial F}{\partial u} - \nabla \cdot \left(\frac{\partial F}{\partial(\nabla u)} \right) = 0.$$

This creates a connection between differential equations, calculus of variations, and optimization.

15.8 Weak Formulation

A weak formulation replaces pointwise differential equations with integral identities against test functions.

For a differential operator $\mathcal{L}$,

$$\mathcal{L}u = f$$

may be converted into an equation of the form

$$\int_{\Omega} (\mathcal{L}u)v dx = \int_{\Omega} fv dx$$

for suitable test functions v .

Weak formulations are particularly important when classical derivatives do not exist.

16 Functional Analysis

16.1 Linear Operators

An operator

$$T : V \rightarrow W$$

is linear if

$$T(ax + by) = aT(x) + bT(y).$$

The operator generalizes the concept of a matrix transformation to potentially infinite-dimensional spaces.

16.2 Operator Norm

For a bounded linear operator,

$$\|T\| = \sup_{x \neq 0} \frac{\|Tx\|}{\|x\|}.$$

Equivalently,

$$\|T\| = \sup_{\|x\|=1} \|Tx\|.$$

16.3 Boundedness and Continuity

A linear operator is bounded if there exists $C > 0$ such that

$$\|Tx\| \leq C\|x\|.$$

For linear operators between normed spaces, boundedness and continuity are equivalent.

16.4 Convergence in Function Spaces

Strong convergence is

$$\|u_n - u\| \rightarrow 0.$$

Weak convergence is written

$$u_n \rightharpoonup u$$

and means

$$\phi(u_n) \rightarrow \phi(u)$$

for every continuous linear functional ϕ .

Strong convergence generally implies weak convergence, but the converse need not hold.

16.5 Hilbert Space Projection

For a closed subspace M of a Hilbert space, every element x can be decomposed as

$$x = P_M x + (x - P_M x),$$

where

$$P_M x \in M$$

and

$$x - P_M x \perp M.$$

This generalizes Euclidean orthogonal projection.

17 Mathematical Foundations of Learning

17.1 Loss Functions

A loss function measures discrepancy between a prediction and a target:

$$\ell(\hat{y}, y).$$

Examples include squared error:

$$\ell(\hat{y}, y) = (\hat{y} - y)^2,$$

and absolute error:

$$\ell(\hat{y}, y) = |\hat{y} - y|.$$

17.2 Risk

For a random pair (X, Y) , the population risk is

$$R(f) = \mathbb{E}[\ell(f(X), Y)].$$

The empirical risk for observations (x_i, y_i) is

$$\hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i).$$

The distinction is fundamental:

$$R(f) \text{ is a population quantity,}$$

while

$$\hat{R}_n(f) \text{ is computed from finite data.}$$

17.3 Empirical Approximation

Under appropriate assumptions,

$$\hat{R}_n(f) \rightarrow R(f)$$

as $n \rightarrow \infty$.

This connects the law of large numbers to statistical learning.

17.4 Generalization

The generalization gap is

$$R(f) - \hat{R}_n(f).$$

A learning procedure is useful when low empirical error corresponds to low population error.

17.5 Overfitting and Underfitting

Overfitting occurs when a function achieves low empirical error but substantially larger population error.

Underfitting occurs when the chosen function class cannot represent the relevant structure sufficiently well.

These phenomena can be viewed mathematically through approximation error and estimation error.

17.6 Regularization

A regularized objective has the form

$$\min_{\theta} \left[\hat{R}_n(\theta) + \lambda \Omega(\theta) \right].$$

Here $\Omega(\theta)$ penalizes certain solutions and $\lambda \geq 0$ controls the strength of the penalty.

Examples include

$$\Omega(\theta) = \|\theta\|_2^2$$

and

$$\Omega(\theta) = \|\theta\|_1.$$

17.7 Bias–Variance Decomposition

For prediction error under squared loss, one can decompose expected error into contributions associated with bias, variance, and irreducible noise.

For an estimator,

$$\text{MSE} = \text{Bias}^2 + \text{Var} + \text{Noise}.$$

More generally, the decomposition explains the trade-off between systematic approximation error and sensitivity to sampled data.

18 Concentration Inequalities

18.1 Markov’s Inequality

For a nonnegative random variable X and $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

18.2 Chebyshev’s Inequality

For a random variable with finite variance,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

18.3 Hoeffding's Inequality

If independent random variables satisfy

$$a_i \leq X_i \leq b_i,$$

then for the sum,

$$\mathbb{P}\left(\sum_i (X_i - \mathbb{E}[X_i]) \geq t\right) \leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right).$$

Hoeffding's inequality gives exponentially decreasing bounds on deviations of bounded independent random variables.

18.4 Chernoff Bounds

For a random variable X , Chernoff bounds use the moment generating function

$$M_X(t) = \mathbb{E}[e^{tX}].$$

For $t > 0$,

$$\mathbb{P}(X \geq a) \leq e^{-ta} \mathbb{E}[e^{tX}].$$

Optimizing over t gives the tightest bound available through this argument.

19 Measure-Theoretic Foundations

19.1 Sigma-Algebras

A sigma-algebra $\mathcal{F}$ on Ω is a collection of subsets satisfying:

$$\Omega \in \mathcal{F},$$

$$A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F},$$

and for a countable collection,

$$A_i \in \mathcal{F} \Rightarrow \bigcup_i A_i \in \mathcal{F}.$$

This provides the mathematical structure needed to define measurable events.

19.2 Measures

A measure μ assigns a nonnegative size to measurable sets:

$$\mu(A) \geq 0.$$

It satisfies

$$\mu(\emptyset) = 0$$

and countable additivity:

$$\mu\left(\bigcup_i A_i\right) = \sum_i \mu(A_i)$$

for pairwise disjoint measurable sets.

Probability is a special case of measure:

$$\mathbb{P}(\Omega) = 1.$$

19.3 Measurable Functions

A function

$$f : \Omega \rightarrow \mathbb{R}$$

is measurable if inverse images of measurable sets are measurable.

Measurability allows integration and expectation to be defined rigorously.

19.4 Lebesgue Integration

The Lebesgue integral generalizes ordinary integration and is particularly suited to limits and measure-theoretic probability.

For a random variable X ,

$$\mathbb{E}[X] = \int_{\Omega} X \, d\mathbb{P}.$$

For a density f_X ,

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f_X(x) \, dx.$$

19.5 Almost Everywhere

A property holds almost everywhere if it fails only on a set of measure zero.

For example,

$$f = g \quad \text{almost everywhere}$$

means

$$\mu(\{x : f(x) \neq g(x)\}) = 0.$$

19.6 Modes of Convergence

Almost sure convergence is written

$$X_n \rightarrow X \quad \text{almost surely.}$$

Convergence in probability is

$$X_n \xrightarrow{P} X.$$

Convergence in distribution is

$$X_n \Rightarrow X.$$

Convergence in L_p is

$$\|X_n - X\|_p \rightarrow 0.$$

These forms of convergence are related but not equivalent in general.

20 Connections Between the Mathematical Foundations

20.1 Linear Algebra and Geometry

Vectors, matrices, inner products, norms, projections, eigenvalues, and singular values provide the geometry of finite-dimensional spaces.

The fundamental relationships are

$$\text{vector space} \rightarrow \text{inner product} \rightarrow \text{norm} \rightarrow \text{distance}.$$

20.2 Calculus and Optimization

Calculus provides local information:

$$\nabla f(x)$$

describes first-order change, while

$$\nabla^2 f(x)$$

describes second-order curvature.

Optimization uses this information to locate minima and maxima.

20.3 Probability and Statistics

Probability describes uncertainty through distributions:

$$P(X).$$

Statistics uses observed samples

$$X_1, \dots, X_n$$

to infer unknown properties of the underlying distribution.

20.4 Probability and Information Theory

Entropy quantifies uncertainty:

$$H(X).$$

KL divergence compares distributions:

$$D_{\text{KL}}(p\|q).$$

Mutual information measures dependence:

$$I(X; Y).$$

20.5 Statistics and Optimization

Maximum likelihood estimation solves

$$\hat{\theta} =_{\theta} L(\theta; x).$$

Equivalently, it minimizes negative log-likelihood:

$$\hat{\theta} =_{\theta} [-\ell(\theta)].$$

Thus statistical inference naturally produces optimization problems.

20.6 Optimization and Convex Analysis

Convexity gives global guarantees.

For a convex function,

$$f(y) \geq f(x) + \nabla f(x)^T(y - x).$$

Consequently, if

$$\nabla f(x^*) = 0,$$

then x^* is a global minimizer.

20.7 Approximation and Function Spaces

Approximation theory asks how well

$$f$$

can be represented by a restricted class.

Functional analysis provides the spaces and norms required to define statements such as

$$\|f - f_n\| \rightarrow 0.$$

20.8 Fourier Analysis and Differential Equations

Fourier analysis converts differential operations into algebraic operations in frequency space.

For example,

$$\widehat{\frac{df}{dx}} = i\omega \hat{f}.$$

This allows many linear differential equations to be transformed into simpler equations in frequency space.

20.9 Probability and Measure Theory

Modern probability is built on measure theory:

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Expectation is integration:

$$\mathbb{E}[X] = \int_{\Omega} X \, d\mathbb{P}.$$

Thus measure theory provides the rigorous mathematical foundation for probability.

21 Core Mathematical Objects

The mathematical foundations can be organized around a small set of central objects.

21.1 Vectors

$$x \in \mathbb{R}^n.$$

Vectors represent points, directions, states, parameters, and finite-dimensional quantities.

21.2 Matrices

$$A \in \mathbb{R}^{m \times n}.$$

Matrices represent linear transformations, systems of equations, quadratic forms, and covariance structures.

21.3 Functions

$$f : X \rightarrow Y.$$

Functions describe mappings between spaces and provide the mathematical language for prediction, approximation, and dynamics.

21.4 Random Variables

$$X : \Omega \rightarrow \mathbb{R}.$$

Random variables represent uncertain quantities.

21.5 Probability Distributions

$$P(X \in A).$$

Distributions describe uncertainty and variability.

21.6 Objectives

$$\min_x f(x).$$

Optimization objectives specify what mathematical quantity is being minimized or maximized.

21.7 Operators

$$T : V \rightarrow W.$$

Operators generalize matrices to function spaces and infinite-dimensional problems.

21.8 Functionals

A functional maps a function to a scalar:

$$J : V \rightarrow \mathbb{R}.$$

Variational problems commonly have the form

$$\min_{u \in V} J[u].$$

22 Summary of the Mathematical Structure

The mathematical foundations can be viewed as a hierarchy:

Sets and Functions $\rightarrow$ Linear Algebra $\rightarrow$ Calculus $\rightarrow$ Probability $\rightarrow$ Statistics

followed by

Information Theory $\rightarrow$ Optimization $\rightarrow$ Convex Analysis

and extended toward continuous and infinite-dimensional mathematics through

Norms $\rightarrow$ Approximation Theory $\rightarrow$ Numerical Mathematics $\rightarrow$ Fourier Analysis $\rightarrow$ Differential Equations $\rightarrow$ Functional Analysis

with probability rigorously grounded in

Measure Theory.

The central mathematical chain is

Data $\rightarrow$ Probability $\rightarrow$ Statistics $\rightarrow$ Objective $\rightarrow$ Optimization $\rightarrow$ Approximation $\rightarrow$ Generalization.

For problems involving physical systems and continuous fields, the chain extends naturally to

Differential Equations $\rightarrow$ Function Spaces $\rightarrow$ Operators $\rightarrow$ Variational Problems $\rightarrow$ Numerical Approximation.